

Literature Status: The ccc Valdivia Dichotomy

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Status: literature already answered (conditional negative answer). The conjecture in arXiv:1505.06727 is explicitly refuted under the Diamond axiom by Theorem 4.1 of the separate paper arXiv:1512.02610. No new mathematical result is claimed here.

1 Original conjecture

C. Correa and D. V. Tausk, *Nontrivial twisted sums of c_0 and $C(K)$* , arXiv:1505.06727, pose the following unnumbered conjecture in Section 4, arXiv PDF page 10 [1].

Source conjecture. *If K is a nonempty Valdivia compact space having the countable chain condition, then either K has a G_δ point or K admits a nontrivial convergent sequence in the complement of a dense Σ -subset.*

The source observes that, under the Continuum Hypothesis, this dichotomy would yield nontrivial twisted sums of c_0 and $C(K)$ for every nonmetrizable Valdivia compact space.

2 Separate later answer

The same authors subsequently wrote *Small Valdivia compacta and trees*, arXiv:1512.02610 [2]. Its abstract and introduction explicitly identify the construction as a counterexample to the conjecture from the source. Theorem 4.1, on arXiv PDF pages 10–11, states:

Later Theorem 4.1. *Assume $\diamond$. There exists a Valdivia compact space K such that*

1. $w(K) = \omega_1$;
2. K has the countable chain condition;
3. K has no G_δ points;
4. K has no nontrivial convergent sequence in the complement of a dense Σ -subset.

Items (2)–(4) are exactly the hypotheses and simultaneous failure of both alternatives in the source conjecture. Thus the later theorem is an explicit conditional counterexample, rather than an implication newly identified by this run.

3 Scope and search status

The answer is conditional on $\diamond$, a principle consistent with ZFC and stronger than CH. Consequently the original unrestricted conjecture cannot be established as a theorem of ZFC, assuming the standard consistency background. The theorem does not provide a counterexample constructed in ZFC. Moreover, the supporting paper states that it does not know whether a nontrivial twisted sum of c_0 and $C(K)$ exists for its counterexample; that Banach-space question remains outside the scope of this status match.

The run registry and local arXiv corpus were searched for the source id, exact conjecture wording, and the terms “Valdivia”, “ccc”, “ G_δ point”, and “convergent sequence”. The decisive match was then checked in arXiv:1512.02610, including its explicit cross-reference and Theorem 4.1.

References

- [1] C. Correa and D. V. Tausk, *Nontrivial twisted sums of c_0 and $C(K)$* , J. Funct. Anal. **270** (2016), 842–853; arXiv:1505.06727.
- [2] C. Correa and D. V. Tausk, *Small Valdivia compacta and trees*, Studia Math. **235** (2016), 117–135; arXiv:1512.02610.